Database Management Systems

CSE Semester 5 — Academic Notes

Synthetic academic knowledge-base document created for CampusAI RAG evaluation and demonstration.

Unit 1 — DBMS Fundamentals

- Database Management System (DBMS) is software used to define, create, store, retrieve, update, and control access to data in databases.
- Advantages of a DBMS include reduced redundancy, improved consistency, data sharing, integrity enforcement, security, concurrency control, backup, and recovery.
- Three-schema architecture separates the database into external, conceptual, and internal levels. External schemas represent user views; the conceptual schema describes the logical structure; the internal schema describes physical storage.
- Data independence is the ability to change one schema level without requiring changes at the next higher level. Physical data independence concerns storage changes; logical data independence concerns changes to the conceptual schema.
- An entity is a distinguishable real-world object. An attribute describes an entity. A relationship represents an association among entities.

Unit 2 — Relational Model and SQL

- The relational model represents data using relations (tables), tuples (rows), and attributes (columns). A domain defines the permitted values of an attribute.
- A super key uniquely identifies a tuple. A candidate key is a minimal super key. A primary key is the candidate key selected as the main identifier. A foreign key references a key in another relation.
- SQL categories include DDL (CREATE, ALTER, DROP), DML (INSERT, UPDATE, DELETE), DQL (SELECT), and transaction/control statements such as COMMIT and ROLLBACK.
- Joins combine rows from related tables. Common types are INNER JOIN, LEFT OUTER JOIN, RIGHT OUTER JOIN, and FULL OUTER JOIN.
- Integrity constraints include domain constraints, key constraints, entity integrity, and referential integrity.

Unit 3 — Functional Dependencies and Normalization

- Normalization is the systematic organization of relations to reduce redundancy and update anomalies while preserving meaningful dependencies.
- A functional dependency $X \rightarrow Y$ means that the value of attribute set X determines the value of attribute set Y.
- First Normal Form (1NF) requires atomic attribute values and no repeating groups. Second Normal Form (2NF) requires 1NF and removal of partial dependencies of non-key attributes on part of a composite key. Third Normal Form (3NF) requires 2NF and removal of transitive dependencies of non-key attributes on a key.
- Boyce-Codd Normal Form (BCNF) requires every determinant of a non-trivial functional dependency to be a candidate key.
- Typical anomalies are insertion anomaly, deletion anomaly, and update anomaly. Decomposition into well-designed relations helps reduce these problems.

- Example: ENROLLMENT(StudentID, StudentName, CourseID, CourseName) can contain repeated student and course information. Decomposition into STUDENT(StudentID, StudentName), COURSE(CourseID, CourseName), and ENROLLMENT(StudentID, CourseID) reduces redundancy.

Unit 4 — Transactions and Concurrency

- A transaction is a logical unit of database work. The ACID properties are Atomicity, Consistency, Isolation, and Durability.
- Atomicity means all operations of a transaction occur or none do. Consistency preserves integrity constraints. Isolation controls interaction among concurrent transactions. Durability preserves committed results after failures.
- Concurrency problems include lost update, dirty read, non-repeatable read, and phantom read.
- Serializability is a correctness criterion for concurrent schedules. Conflict serializability can be analyzed using a precedence graph.
- Lock-based protocols use shared locks for reading and exclusive locks for writing. Two-phase locking has a growing phase followed by a shrinking phase.

Unit 5 — Recovery and Indexing

- Database recovery restores the database to a consistent state after failures such as transaction errors, system crashes, or storage failures.
- Write-ahead logging (WAL) requires relevant log records to be written before modified data pages are written to stable storage.
- Checkpoints reduce recovery work by recording a known point from which recovery can proceed.
- An index is an auxiliary data structure that speeds up data retrieval at the cost of additional storage and update overhead.
- B+ trees are widely used for database indexing because their balanced structure supports efficient search and range queries.

Quick Revision Checklist

- Know the key definitions and distinguish closely related concepts.
- Practice explaining each concept in your own words before using it in an exam.
- For numerical or algorithmic topics, work through at least one example by hand.
- For database and software-engineering topics, connect definitions to practical examples.

CampusAI RAG metadata suggestion: department=CSE, semester=5, subject=Database Management Systems